

AAYUSH RAI

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EDUCATION

Northeastern University

Master of Science in Computer Science

Expected May 2028

Boston, MA

Kansas State University

Bachelor of Science in Computer Science and Math Dual Major

August 2026

Manhattan, KS

- **Relevant Coursework:** Data Structures and Algorithms, Object Oriented Programming, Prob & Stat, Database Systems, Operating Systems, Computer Architecture, Software Design

SKILLS

- Python, C#, SQL, JavaScript, React.js, Node.js, Gradio, Streamlit, FastAPI, OpenCV, YOLOv11
- Pandas, NumPy, Scikit-learn TensorFlow, PyTorch, Git, ROS, REST APIs, Linux

EXPERIENCE

Machine Learning Researcher | ARISE, Kansas NSF

Nov 2025 – July 2026

- Engineered a RAG pipeline using LangChain, ChromaDB and transformer-based embedding to semantically index and query large-scale emergency response documents achieving 90%+ retrieval precision.
- Architected an interface using Gradio and FastAPI with asynchronous Python orchestration to handle concurrent stakeholder queries, maintaining less than 10 sec response latency across REST API calls on a Linux deployment environment.

Computer Vision Researcher | CNAP, Dept. of Computer Science

Feb 2025 – May 2025

- Developed and trained computer vision pipelines using YOLOv11, TensorFlow and DeepLabCut on high-volume datasets to improve pose estimation accuracy resulting in higher mean Average Precision across validation datasets.
- Automated end-to-end data preprocessing using Python, OpenCV, Pandas, NumPy and JSON to CSV transformation pipelines so that raw experimental data could be standardized resulting in a 40% acceleration in dataset labeling.

PROJECTS

LLMGuard | AI Model Attestation

Spring 26

- Formulated an attestation framework with TPM and IMA file integrity with TEE-backed behavioral attestation for runtime memory integrity, defending locally deployed transformer LLMs against 4 distinct classes of integrity attacks on Linux.
- Modeled the end-to-end threat surface demonstrating that complementary defense layers are required to mitigate identified attack vectors spanning disk-level tampering and microarchitectural memory exploits.

BookFlix | Full Stack Digital Library System

Fall 25

- Architected a normalized SQL Server schema with indexing and ACID-compliant transactions to manage 2,000+ records, ensuring data integrity for concurrent user reviews and authentication.
- Developed backend services in ASP.NET Core (C#) using direct SQL execution to ensure reproducible query behavior and integrated a React, TypeScript frontend enabling search, reviews, user authentication, and dynamic book discovery.

AguaCrop | Kansas State TAPS Hackathon Winner

Fall 24

- Designed and implemented a hyper-trained RAG-based chatbot using Python and natural language processing techniques to deliver interactive, real-time irrigation insights leveraging Node.js for backend API and server-side data processing.
- Built predictive models using Scikit-learn on historical agricultural and weather data and integrated output with a React.js frontend via REST APIs to visualize irrigation schedules, delivering a functional MVP within a 48-hour sprint.